

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 3 Winter 2025 (Supply.)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 25AF1293PC302 - ELECTRICAL AND ELECTRONICS MEASUREMENTS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1. A PMMC instrument has a full-scale deflection current of $50 \mu\text{A}$ and internal resistance of 1000Ω . What shunt resistance is required to extend its range to 5 A ?
a) 0.01Ω b) 0.02Ω c) 0.05Ω d) 0.1Ω
2. A PMMC voltmeter reads 200 V using a series resistance of $1.99 \text{ M}\Omega$. If the full-scale deflection current is $100 \mu\text{A}$, what is the internal resistance of the meter?
a) 100Ω b) 200Ω c) 1000Ω d) $10 \text{ k}\Omega$
3. A single-phase load takes 10 A at 230 V with a power factor of 0.8 lagging. What is the true power consumed?
a) 1.84 kW b) 2.30 kW c) 1.50 kW d) 3.20 kW
4. An energy meter makes 600 revolutions per kWh. How many revolutions will it make when 2.5 kWh energy is consumed?
a) 1200 b) 1500 c) 1800 d) 2000
5. In a Wheatstone bridge, the ratio arms are $P = 100 \Omega$ and $Q = 1000 \Omega$. The standard resistance $R = 10 \Omega$. Find the unknown resistance S at balance.
a) 1Ω b) 10Ω c) 100Ω d) 1000Ω
6. A Kelvin double bridge is used to measure a resistance of 0.01Ω . If the ratio arms are equal, the error due to lead resistance is:
a) Maximum b) Minimum c) Equal to lead resistance d) Infinite
7. A current transformer has a ratio of $500/5 \text{ A}$. If the primary current is 400 A , what is the secondary current?
a) 2 A b) 4 A c) 5 A d) 8 A <https://www.batuonline.com>
8. A platinum RTD has resistance 100Ω at 0°C . If temperature coefficient is $0.00385/^\circ\text{C}$, find resistance at 100°C .
a) 110Ω b) 120Ω c) 138.5Ω d) 150Ω
9. An LVDT has sensitivity of 5 mV/mm . If the output voltage is 25 mV , find the displacement.
a) 2 mm b) 3 mm c) 5 mm d) 10 mm